

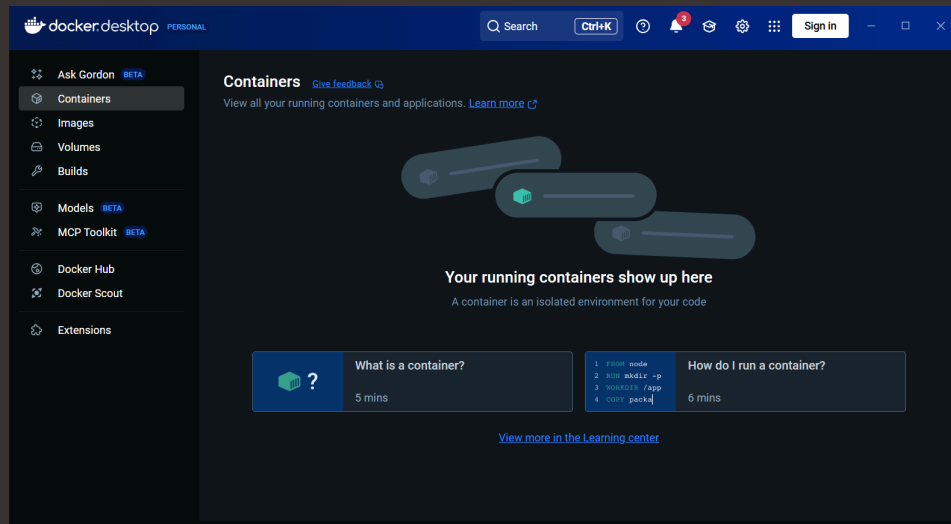


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Deploy an App with Docker



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Introducing Today's Project!

What is Docker?

Docker is a containerization platform which allows apps to be packaged together with their dependencies and run in isolated environments. It is useful because it makes applications easy to deploy and ensures they work the same on different systems. I used Docker in this project to deploy my web application in a simple and consistent environment.

One thing I didn't expect...

One thing I didn't expect in this project was the versatility of docker and AWS Elastic Beanstalk.

This project took me...

This project took me approximately 50 minutes.



Understanding Containers and Docker

Containers

Containers are a way to package up applications, dependencies, libraries and also settings into a lightweight unit, which helps it run smoothly. They are useful because they allow applications to run the same way on any system, make deployment easier, and reduce issues caused by differences in operating environments

A container image is a blueprint or template that contains everything needed to run an application, including the application code, runtime, libraries, dependencies, and other configuration files.

Docker

Docker allows for the creation, running, and managing of containers, which makes it easy to package applications with everything they need to work the same way on any system. Docker Desktop is the application that makes Docker easy to use on a personal computer, by providing a simple interface and all the tools needed to build and run containers on different OS.



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The Docker daemon is a background service that is responsible for managing Docker objects such as the containers, images, networks, and volumes. It listens for commands from the Docker client and executes operations required to build, run, and manage containers.



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Running an Nginx Image

Nginx is a web server, nginx is a program to run websites and web apps. It is designed to efficiently handle a large number of simultaneous connections while using minimal system resources.

The command I ran to start a new container was `docker run -d -p 80:80 nginx`

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.



Creating a Custom Image

The Dockerfile is a file, a text file that contains a set of instructions used by Docker in building a container image, defining how the application and its environment are set up.

My Dockerfile tells Docker three things, the first is that the image should be a template from an official Nginx image as the base image for the container, it also tells Docker to copy index.html file into the default directory where Nginx serves web content and finally it tells the Docker that the container listens on port 80 (the standard HTTP port).

The command I used to build a custom image with my Dockerfile was `Dockerfile -t my-web-app`. The '.' at the end of the command means that the dockerfile should find the Dockerfile in the Compute directory.

```
Terminal

Abdullahi Yakubu@DESKTOP-08HCGPJ MINGW64 ~/Desktop/Compute
$ cat Dockerfile
FROM nginx:latest
COPY index.html /usr/share/nginx/html/
EXPOSE 80

Abdullahi Yakubu@DESKTOP-08HCGPJ MINGW64 ~/Desktop/Compute
$
```



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Running My Custom Image

I ran into an error when i tried running my custom image on port 80, this error occurred because there was an image running on the port 80. I resolved this by looking into the running processes and stopped the image.

In this example, the container image is my-web-app.

Hello from vim's custom Docker image!



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Elastic Beanstalk

Elastic Beanstalk is a service provided by AWS which allows the deployment of cloud apps without worrying about the underlying infrastructure. It handles things like capacity provisioning, Load balancing, Auto-scaling and health monitoring.

Deploying my custom image with Elastic Beanstalk took me forty minutes, which was interesting learning and reading through the write ups..

Hello from vim's custom Docker image!

If I can see this, it means Elastic Beanstalk has deployed an image with my work



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Deploying App Updates

To learn how to deploy app updates with Elastic Beanstalk, I updated my app by adding an img tag and other attributes. I verified those changes by opening the html file in a browser.

My app updates didn't show up in my live environment immediately because I only made the changes locally so the changes wouldnt reflect. To deploy my changes, I only had to get to my environment page then clicked on upload and deploy. Voila!!!

Hello from vim's custom Docker image!

If I can see this, it means Elastic Beanstalk has deployed an image with my work

And here's a special image chosen by me:





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